

National University of Computer and Emerging Sciences



Lab Exercise 13 AL2002-Artificial Intelligence Lab

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Exercise

1. Clustering using built in libraries
 - Use the code provided in the attached lab manual to implement **k-means** and **k-medoids** using built in libraries on the given dataset.
 - Use the elbow method to identify the optimal value of k.
2. k-means from scratch
 - Complete the following functions in the partial implementation of k-means provided in the attached .ipynb file.
 - `assign_clusters`
 - `k_means`
 - Complete `update_centroids` function to find the average of all points in this cluster to get the new center (centroid)
3. k-medoids using built in libraries
 - Modify your k-means implementation to implement k-medoids from scratch.
 - Compare the result of k means and k medoids. Which one performs better?